# Sanjay Mehta

AI Research Scientist & Systems Architect

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## Summary

Passionate researcher and software engineer specializing in Large Language Model Alignment, Efficient Neural Architectures, and High-Performance Distributed Systems. Dedicated to building open-source tools and publishing peer-reviewed research.

## Education

### Ph.D. in Computer Science & Artificial Intelligence

Massachusetts Institute of Technology (MIT) | 2019 – 2022

Dissertation: Efficient Sub-quadratic Transformer Attention Mechanisms

### B.S. in Computer Science & Mathematics

Stanford University | 2015 – 2019

First Class Honors, GPA 3.96 / 4.00

## Professional Experience

### Senior AI Research Scientist — Apex AI Research Labs

2024 – Present | San Francisco, CA

Leading research on efficient inference acceleration for 70B+ parameter LLMs. Optimizing KV-cache memory footprints and sub-quadratic attention kernels.

- Reduced inference latency by 42% on H100 clusters.
- Published 2 top-tier papers at NeurIPS & ICML.
- Mentored 4 PhD research interns.

### Machine Learning Systems Engineer — ScaleTech Solutions

2022 – 2024 | Boston, MA

Designed and deployed real-time recommendation engines processing over 10M requests daily using Rust and PyTorch C++ API.

- Architected distributed pipeline reducing p99 latency from 180ms to 24ms.
- Awarded Engineering Innovation Award 2023.

## Technical Skills

**Artificial Intelligence:** PyTorch, Transformers, vLLM, RLHF / DPO, CUDA, LangChain

**Systems & Cloud:** Python, Rust, C++, Docker, Kubernetes, Ray

**Web & Front-end:** React, TypeScript, Vite, Node.js, GraphQL, Tailwind/Vanilla CSS

**Tools & Methodologies:** Git, LaTeX, Weights & Biases, Linux, CI/CD, Distributed Training